Week 12, Day 1 — presenter notes

Slides: `week-12-day-01.html` → **F** · arrows · **N** · **P**.

Mutex, WaitGroup, context, races. Lab = `-race` red → mutex → green. Context signatures + timeout demo + graceful shutdown sketch.

Timing (3–3.5 hours)

Clock	Block	Slides
0:00–0:15	Week 11 check	1–2
0:15–0:30	Outcomes + SE	3–4
0:30–1:30	Race, Mutex, RWMutex, WaitGroup	5–8
1:30–2:30	Context, Gin, SMS, shutdown	9–12
2:30–end	Lab red→green + demos	13–end

Never skip the race FAIL on the projector.

Live demo

1. `go test -race ./counter/` FAIL → add Mutex → PASS
2. Timeout demo endpoint
3. Optional Ctrl+C graceful shutdown

Lab gate

- [ ] `-race` green on race-lab
- [ ] Mutex vs WaitGroup explained
- [ ] Repo takes `context.Context`; handlers pass `c.Request.Context()`

Homework

README documents `go test -race ./...`. PR explains what was red / what fixed it.

What not to teach

Postgres cancel, deep testing/CI (week 13), “replace every mutex with a channel.”